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SPECIAL ARTICLE

Linear regression analysis for comparing two measurers or methods of measurement: But which regression?

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SUMMARY

1. There are two reasons for wanting to compare measurers or methods of measurement. One is to calibrate one method or measurer against another; the other is to detect bias. Fixed bias is present when one method gives higher (or lower) values across the whole range of measurement. Proportional bias is present when one method gives values that diverge progressively from those of the other.

2. Linear regression analysis is a popular method for comparing methods of measurement, but the familiar ordinary least squares (OLS) method is rarely acceptable. The OLS method requires that the *x* values are fixed by the design of the study, whereas it is usual that both *y* and *x* values are free to vary and are subject to error. In this case, special regression techniques must be used.

3. Clinical chemists favour techniques such as major axis regression ('Deming's method'), the Passing–Bablok method or the bivariate least median squares method. Other disciplines, such as allometry, astronomy, biology, econometrics, fisheries research, genetics, geology, physics and sports science, have their own preferences.

4. Many Monte Carlo simulations have been performed to try to decide which technique is best, but the results are almost uninterpretable.

5. I suggest that pharmacologists and physiologists should use ordinary least products regression analysis (geometric mean regression, reduced major axis regression): it is versatile, can be used for calibration or to detect bias and can be executed by hand-held calculator or by using the loss function in popular, general-purpose, statistical software.

Key words: Altman–Bland, bias, calibration, confidence interval, loss function, non-parametric methods, outlying values.

INTRODUCTION

In 1977, Robert Brace wrote an article in the *American Journal of Physiology* on techniques for comparing two methods of measurement.[1] It has been cited nearly 140 times, chiefly by physiologists. Twenty years later, I wrote on the same topic,[2] although my paper has attracted only 94 citations. Both Brace and I commended reduced major axis regression analysis. What has happened since to persuade me to revisit the topic? Some of the reasons are listed below.

1. Altman and Bland's proposal to use the method of differences,[3–6] their *Lancet* article having been cited nearly 16 000 times.
2. Intense activity among allometrists, astronomers, biologists, economists, fisheries researchers, geneticists, geologists, physicists, sports scientists and, especially, clinical chemists. In each of these disciplines there seems to be a favourite technique of linear regression analysis. I single out clinical chemists because, of the more than 55 articles I have consulted on linear regression techniques for comparing methods of measurement, no fewer than 25 (45%) were written by or for chemists. In only a handful of these articles did reduced major axis regression find favour.[7–10] So, was the advice that Brace and I gave to physiologists and pharmacologists misleading or even plain wrong?
3. The extensive use of Monte Carlo simulations in attempts to identify the best method of regression analysis.

I shall address these three matters in turn but, first, I had better remind readers of the distinction between Model I and Model II regression, as defined by Sokal and Rohlf.[11]

The familiar form of linear regression analysis is known variously as ordinary least squares (OLS) or simple linear regression. It falls within the category of Model I regression analysis.[11] The coefficients in the regression model $Y_{est} = a + bx$ are calculated by minimizing the sum of the squares of the deviations of the *y* values from the line of best fit (Fig. 1a,b). However, it is a requirement for OLS regression analysis that the *x* values are fixed by the design of the experiment and are not subject to error.[1,10–14] This is an uncommon occurrence when biological methods of measurement are compared. Some authors admit of circumstances in which Model I regression analysis could be used when the *x* values are not fixed. One is if there were a perfect correlation between *y* and *x* ($r = 1.00$). If $r = 1.00$, then the lines $Y_{est} = a + bx$ and $X_{est} = a + by$ are identical; if $r < 1.00$, the slope (b) of the former is less than that of the latter (Fig. 2a). Others have argued that Model I regression analysis is permissible if the values of *x* are not fixed, provided that it would be

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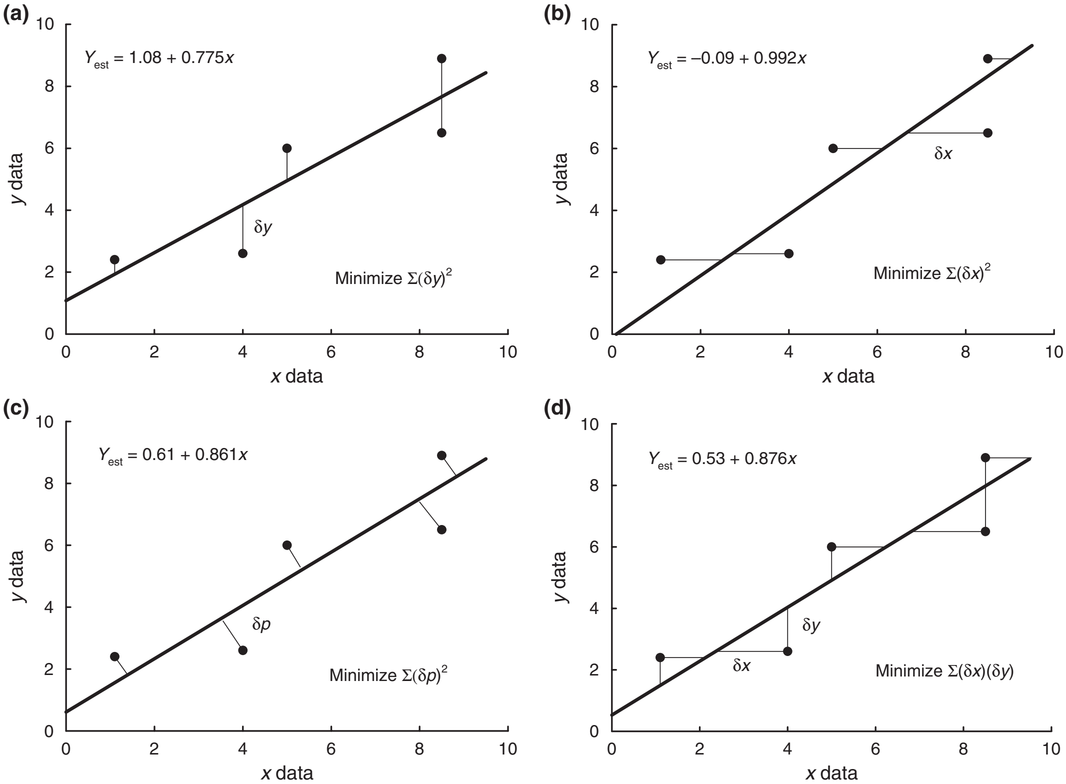

Fig. 1 Techniques for linear regression analysis in method comparison. The data set used is available in Table S2a. (a) Ordinary least squares of y regression ($OLS_{y.x}$); $\Sigma(\delta y)^2$ is minimized. (b) Ordinary least squares of x regression ($OLS_{x.y}$); $\Sigma(\delta x)^2$ is minimized. (c) Major axis (least perpendicular squares) regression (MA); $\Sigma(\delta p)^2$ is minimized. (d) Ordinary least products regression (OLP); $\Sigma(\delta x)(\delta y)$ is minimized.

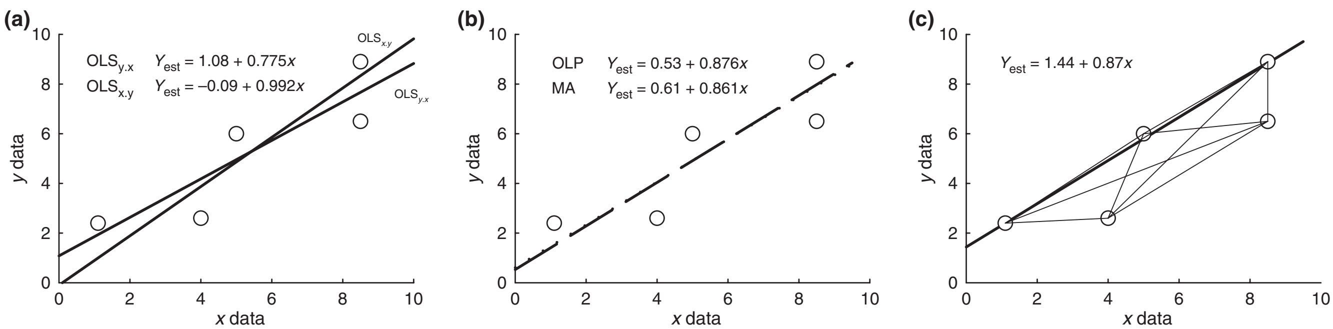

Fig. 2 Worked examples of techniques for linear regression analysis in method comparison. Data set from Ludbrook[29] and available in Table S2. (a) Ordinary least squares of y regression ($OLS_{y.x}$) and x regression ($OLS_{x.y}$) lines. Note that the slope of the $OLS_{y.x}$ line is always less than that of the $OLS_{x.y}$ line and the two lines intercept at $\bar{x}, \bar{y}$. (b) Ordinary least products regression (OLP; - -) and major axis (least perpendicular squares) regression (MA; - - - -) lines. Note that because b is close to 1, the lines are nearly identical. (c) Passing–Bablok rank-order regression line. Note that its slope is the median of all $n(n - 1) = 20$ lines joining the pairs of x,y points (omitting $b = 0$ or ∞). The Passing–Bablok line passes through, or close to, the median values of x and y.

absurd to think of predicting x from y rather than y from x.[11,13] Examples are the prediction of children's heights from those of their parents or the prediction of blood pressure from body mass index.

When the values of x are not fixed, a form of Model II regression analysis should be used.[11] The trouble is that a very large number of forms of Model II linear regression analysis have been described (see Table S1 available as Supporting Information for this paper). The purpose of the present article is to try to decide which is best for physiologists and pharmacologists.

I interpolate here a short note on error. Two varieties can be identified. One results from natural or biological variation. Because of this, the x,y values in a sample can never sit exactly on a straight line. The other variety is measurement error. This comes about because no method of measurement is perfect, whether because of inconsistencies in behaviour of the measuring instrument or inconsistency on the part of those using it. The overall error can be estimated by analysis of the residuals from the regression. The measurement error can be estimated by way of replicate measurements. Statisticians make much of this distinction between sources of error, but it is relatively unimportant in the context of the present review.

There are two distinct reasons for wanting to compare methods of measurement. The first is when one wishes to compare a new method with an existing one to see whether the results from the two methods are indistinguishable or whether they differ in an important way. The second reason is to calibrate one method against another, so that measurements made by one method can be converted into those made by the other.

Investigators in different disciplines have their favourite techniques for comparing methods of measurement. Clinicians prefer the method of differences, presented by Altman and Bland in several articles that appeared in wide-circulation general and specialized medical journals.[3–6] Their technique has the supposed merit of being simple to execute, although this is not always true.[15] One purpose of the method of differences is to expose bias between methods but, as I and some others have demonstrated,[2,15,16] it does not always do so accurately. As Bland and Altman refined their method, they focused attention on what they called 'limits of agreement',[6] which are 95% confidence limits for the differences. They postulated that if the limits of agreement were wide, then it would be wrong for clinicians to substitute one method for the other. Other critics have found fault with the ways in which the method of differences is interpreted, especially by clinical chemists.[17] Moreover, it is unsuited to calibration so, if only for this reason, I shall not consider it further.

Clinical chemists are concerned chiefly with calibrating one method against another. Their favourite technique seems to be least squared perpendicular distance regression analysis (also known as major axis (MA) regression and Deming's method), in which the squares of the perpendicular distances of the x,y points from the regression line are minimized (Figs 1c,2b).[2,18–23] A variant is weighted major axis regression (WMA), used if there is increasing scatter of the y values as x increases (heteroscedasticity). However, some clinical chemists favour the Passing–Bablok (P-B) technique, in which the slopes of the lines interconnecting all possible pairs of data points are calculated.[24–26] The median slope is taken to be that of the best-fitting line (Fig. 2c). A few clinical chemists have a penchant for the bivariate least medians squared (BLMS) method.[27,28] This is a form of robust regression (see Ludbrook[29]), extended from the regression of y on x to include that of x on y. It is argued that it is particularly well suited to x,y data sets in which there are outliers. But this is only a beginning: I have been able to identify over 20 different forms of Model II regression analysis (see Table S1).

Almost the only point on which all parties agree is that the Pearson production–moment correlation coefficient (r) is unsuitable for detecting bias. It is merely an index of linear association between y and x values.

I shall proceed by reminding readers of what is meant by bias, then describe some of the techniques of regression favoured by different disciplines, including least products regression. I shall also present the results of 13 Monte Carlo simulation studies, because this is one way in which theoretical results can be tested.

BIAS AND ITS DETECTION

Whatever form of Model II linear regression is used, it is usually not good enough merely to estimate the regression coefficients, although this may suffice if the goal is calibration of one method against another. One usually also wants to test for the existence of bias. This can take one, or both, of two forms. If there is fixed bias, then one method gives values that are consistently greater (or smaller) than the other, across the whole range. If there is proportional bias, then one method gives values of y that are progressively greater (or smaller) than the other, in proportion to the values of x. Whereas the regression coefficients can be sometimes estimated with a hand-held calculator or spreadsheet, the determination of bias requires using complex, iterative algorithms in a computer program so as to estimate the standard errors of the coefficients. This, in turn, allows the calculation of 95% confidence intervals (CI) for a and b. There are other methods for calculating 95% CIs. One is bootstrapping[2,29] and Manly suggests a randomization (permutation) approach,[30,31] but this does not seem to have been used in method comparison studies. If the 95% CI for a does not include 0, then one can infer that there is fixed bias. If the 95% CI for b does not include 1.0, then one can infer that there is proportional bias. This is provided, of course, that the measurements made by the two methods are on the same scale.

TECHNIQUES FOR MODEL II REGRESSION ANALYSIS

There are three procedures for comparing methods of measurement that seem to be especially popular among clinical chemists. These are MA regression (least squared perpendicular distance linear regression analysis, Deming's method), the P-B rank-order technique and the BLMS technique.

Major axis regression (squared least perpendicular distance regression, Deming's method)

Clinical chemists prefer to call this 'Deming's method', although Adcock described it in 1878,[18] Karl Pearson in 1901[23] and Ragnar Frisch in 1929.[32] H Edwards Deming was a remarkable man who started off as a statistician/mathematician in the US Department of Agriculture and ultimately became deservedly famous for enunciating the principles of quality control and marketing in the manufacturing industry. In the late 1930s, he addressed the problem of estimating regression lines when both x and y variables are free to vary and subject to error.[19–21] Deming's solution, like that of his predecessors, was to minimize the sum of the squares of the perpendicular distances of the data points from the straight line of best fit (Fig. 1c). There is no simple, pencil-and-paper way of performing MA regression analysis, but it can be executed by using the loss function within the non-linear modules of general statistical software such as SYSTAT, SPSS/PASW, or SAS and by purpose-designed software (see Appendix S1).

A word on 'loss function'. This term seems to have been borrowed by statisticians from the disciplines of economics and manufacturing. In the present context it refers to expressions such as those in Appendix S2. The loss is, in effect, the sums of the squared differences between observed and predicted values, which are minimized by an iterative computational process to arrive at best estimates for slope and y intercept.

There are certain theoretical requirements that must be met if the Deming technique is to be accurate. These are: (i) the scales of the x and y variables must be identical; (ii) the 'true' line for the population, $Y = \alpha + \beta X$, must pass though 0, 0; (iii) the slope of the 'true' population line, β, must be close to 1.0; and (iv) the correlation coefficient for the population, ρ, must be close to 1.0. Obviously, not many data sets fulfil all these requirements.

Standardized major axis regression

The requirements for MA referred to above are rarely met in practice. This has led to the development of standardized major axis regression (SMA).[1,9,33,34] The process of standardization is designed to equalize the x and y scales of measurement when these are, for instance, in different units. It usually takes the form of multiplying the y values by s_x/s_y, where s_x and s_y are the standard deviations of x and y, respectively. Major axis regression is then conducted in the usual way and the regression coefficients back-transformed by dividing them by s_x/s_y. The outcome of this rather tedious process is that the regression coefficients are identical to those obtained by ordinary least products (OLP) regression.

Weighted major axis regression

A further, not uncommon, problem is that the scatter (variance) of the y values increases in parallel with the increase in x values (heteroscedasticity). This can be corrected for by weighting.[2] Weighted major axis regression (WMA) can be executed by choosing a suitable loss function in SYSTAT (Systat Software Inc., Chicago, IL, USA), SPSS/PASW (SPSS Inc., Chicago, IL, USA) or SAS (SAS Institute Inc., Cary, NC, USA) (see Appendix S2) or by using Analyse-it (Analyse-it Software Ltd., Leeds, UK) (see Appendix S1).

Least products regression analysis

Least products regression analysis has many synonyms (Table S1). It was described by Woolley under the title 'minimized area regression',[10] at about the same time as Deming rediscovered the perpendicular distance method.[19] However, the Nobel prize-winning economist Samuelson[35] correctly gives precedence to Ragnar Frisch (also a Nobel prize-winning economist).[31] The technique has been reinvented many times under one or another name.[2,7,20,21,35–37] The principle underlying the technique is to minimize the sum of the products of the x and y deviations from the line of best fit (hence OLP) or, and it is exactly the same, to minimize the absolute areas of the rectangles whose sides are formed by the x and y deviations from the line of best fit.[10]

There are three very simple methods for estimating the coefficients for an OLP regression $Y_{est} = a + bx$. The formulae for the slope, b, of the regression are as follows:

$$b_{y.x}^{OLP} = s_y/s_x \quad [1]$$

where $b_{y.x}^{OLP}$ is the slope of the OLP regression of y on x, s_y and s_x are the standard deviations of the y and x values, respectively, and the appropriate sign is attached to the slope;

$$b_{y.x}^{OLP} = b_{y.x}^{OLS}/r \quad [2]$$

where $b_{y.x}^{OLS}$ is the slope of the OLS regression of y on x and r is Pearson's product–moment correlation coefficient; and

$$b_{y.x}^{OLP} = \sqrt{(b_{y.x}^{OLS})(1/b_{x.y}^{OLS})} \quad [3]$$

where $b_{x.y}^{OLP}$ is the slope of the OLP regression of y on x, $b_{x.y}^{OLS}$ is the slope of the OLS regression of y on x and $b_{x.y}^{OLS}$ is the slope of the OLS regression of x on y. This formula accounts for the synonym 'geometric mean regression'.

Incidentally, the formula for Pearson's product–moment correlation coefficient is:

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}} \quad [4]$$

Note that, as in OLP regression, r takes into account the deviations of both x and y from their respective means.

The y intercept for the regression, a, is obtained from the regression model:

$$\bar{y} = a + b\bar{x} \text{ or, rearranging, } a = \bar{y} - b\bar{x} \quad [5]$$

where $\bar{x}$ and $\bar{y}$ are the means of x and y, respectively.

It is a property of an OLP regression line, shared by the $OLS_{y.x}$, $OLS_{x.y}$ and the MA regression lines, that it passes through the points $\bar{x}, \bar{y}$ (Fig. 2a,b).

Passing–Bablok rank-order linear regression analysis technique

The P-B rank-order linear regression technique was described in the early 1980s.[24–26] The logic behind it is not easy to follow, although it has something of the flavour of bootstrapping.[30] All possible pairs of x,y points are connected and the slopes of the resulting connecting lines are placed in rank order. For n x,y points, the number of possible lines is $n(n-1)/2$. The median value of the slopes is taken to be that of the line of best fit. Passing and Bablok recommended excluding from this process lines in which $b = 0$ or $b = \infty$, although this practice is not always followed. The y intercept is calculated, by analogy with eqn [5], as:

$$a = \tilde{y} - b\tilde{x} \quad [6]$$

where $\tilde{x}$ and $\tilde{y}$ are medians.

It is clear that the procedure falls into the category of being non-parametric. One of its supposed advantages is that it copes very well with outlying values.[24,25] The P-B procedure can be executed by means of specialized software or, more laboriously, by way of a spreadsheet (see Appendix S1).

I used the spreadsheet technique to check on the specialized program Answer-it for Excel, although on a very small data set in which $n = 5$ (Fig. 2c). The two methods agreed quite well, although I inferred that Answer-it does not exclude slopes of ∞ or 0. The result was, however, completely at odds with the results of using the MA or OLP methods (Fig. 2). This was presumably because of the very small sample. Passing and Bablok suggest that n should be > 50.[25]

Bivariate least medians squared

The BLMS method is an extension of the more familiar method of univariate least medians squared (LMS; see Ludbrook[29] and Teissier[37]), a form of robust regression that has been recommended when there are one or more outlying x, y values. The BLMS method is the brainchild of Feldmann *et al.*[27] and Del Rio *et al.*[28] As with the P-B technique, BLMS involves interconnecting all possible $n(n-1)/2$ pairs of points.[28] It results in equal numbers of x, y points above and below the BLMS line of best fit.[28] None of the general or specialized statistics programs in Appendix S1 provides for BLMS.

Other techniques

A host of other techniques have been described, many of which are listed in Table S1.

Parametric (OLP/MA) versus non-parametric (P-B/BLMS)

In order to try to grasp the difference between parametric (OLP, MA) and non-parametric (P-B, BLMS) techniques I compared the outcome of regression analysis by all four methods on an artificial set of data that I have used previously (Table S3).[29] The outcomes are shown in Fig. 3. When there was no outlier, the four regression lines were almost identical (Fig. 3a). When there was an outlier, whether 'off line' (Fig. 3b) or 'on line' but extreme and exerting considerable leverage (Fig. 3c), the OLP and MA regression lines were distinctly different from the P-B and BLMS lines. I repeated these analyses on the real-life set of data from Rousseeuw and Leroy,[38] reproduced in Table S3. The outcomes are in Fig. 4 and resemble closely those of Fig. 3b. If the three outliers in Table S3 and Fig. 4 are deleted, the OLP, MA, P-B and BLMS regression lines are almost identical. In summary, it is almost as though the P-B and BLMS techniques ignore outliers, a practice to be strongly discouraged.[29,39]

Monte Carlo simulation studies

The process of Monte Carlo simulation is started by creating a computer-generated 'population' that conforms to one or another mathematically defined frequency distribution. In the present context, this usually means a bivariate normally distributed population of X,Y values, although other bivariate distributions, such as log-normal or uniform, are sometimes used and sometimes outlying values are added. The next step is to take random (more accurately, pseudorandom) samples from the population. These are then subjected to the statistical test(s) of interest, in this case one or more forms of Model II regression analysis.

The Monte Carlo approach allows the bivariate population of X, Y values to be specified not merely in terms of its frequency distribution, but also in terms

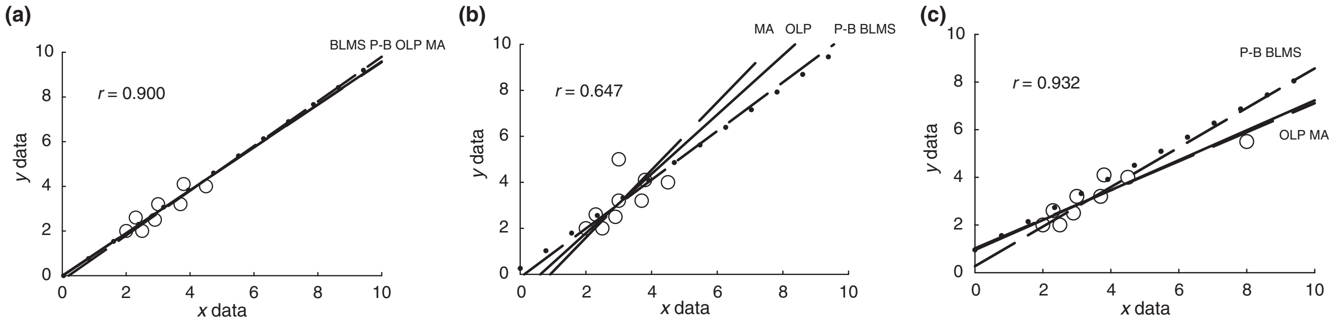

Fig. 3 Ordinary least products (OLP; —), major axis (MA; – – – –), Passing–Bablok (P-B; - - - -) and bivariate least median squares (BLMS; · · · ·) regression lines. (a) No outliers or leverage. (b) A single outlier. (c) Excessive leverage (data set from Ludbrook[29] and in Table S2). *r*, Pearson's correlation coefficient.

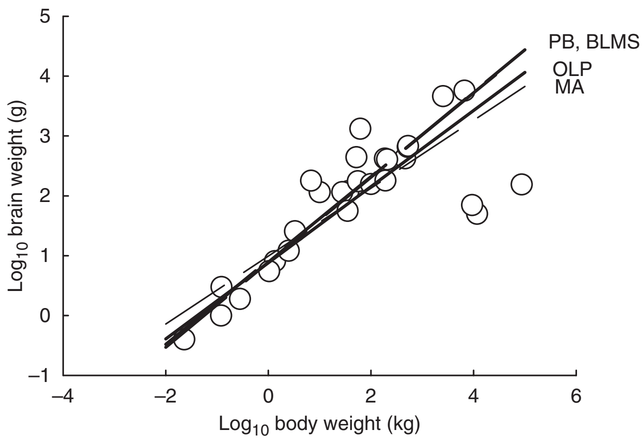

Fig. 4 Ordinary least products (OLP; —), major axis (MA; – – – –), Passing–Bablok (P-B; - - - -) and bivariate least median squares (BLMS; · · · ·) regression lines. Data from Roussuuuw and Leroy[37] and reproduced in Table S3. *r*, Pearson's correlation coefficient.

of the coefficients α and β for the 'true' linear regression model $Y = \alpha + \beta X$ and in terms of the 'goodness-of-fit' of the model in terms of the population correlation coefficient (ρ). The Monte Carlo process allows the size of the random samples of *x*,*y* values to be specified (e.g. $n = 10$ or $n = 100$) and the number of samples taken for study (e.g. 100–1000).

The outcome of the Monte Carlo procedure can be expressed as the sample regression coefficients, *a* and *b*, for the regression $Y_{est} = a + bx$. It can also be expressed as the presence or absence of 'significant' bias, fixed or proportional (see above). The goal is to see which forms of Model II regression analysis give sample regression coefficients that are closest to those for the population and are associated with least bias.

In Table 1, I have listed 13 Monte Carlo simulation studies. Seven originated from chemistry departments and one each came from allometry, astronomy, biology, geology, mathematics and pharmacology. In nine of the 13 studies, β was set at, or close to, 1.0. In only five was ρ specified, ranging from 0.1 to 1.0. Sample sizes ranged from $n = 6$ to $n = 100$. Ten studies tested MA regression, four tested OLP or SMA and one each tested the P-B rank-order and BLMS techniques.

What conclusions can be drawn from these studies? It may be mere coincidence, but in the studies that favour the OLP technique samples were taken from populations with a wide range of correlation coefficients ($\rho = 0.1$–0.75).[40,41] It again may be coincidental, but in the studies that favour the MA technique samples were taken from populations in which β was close to 1.0.[34,42] In the only study that favours the P-B technique, samples were taken from populations in which there was heteroscedasticity.[22] In the only study of BLMS, outliers were introduced into the bivariate normal population.[27] It is scarcely surprising that when the sampled populations exhibited heteroscedasticity, the results of weighted regressions (WMA in particular) were said to be superior to unadjusted regressions.[22,32,34]

To my mind, there is a serious logical flaw in the Monte Carlo simulation approach. The Monte Carlo method implies that biomedical investigators take samples randomly from defined parent populations. In real life, we take samples of convenience and randomize them to one or another 'treatment'. In the present context, for 'treatments' read 'methods of measurement'. So, the statistical inference from method-comparison studies refers only to the study that was actually performed, not to some bivariate population of values that was randomly sampled.

DISCUSSION AND RECOMMENDATIONS

Would it not have been easier if I had referred readers to standard, readily available, statistical texts? Maybe, but the standard biostatistical texts barely mention the distinction between Model I and Model II regression analyses and only give substantial attention to Wald's two-sample and Bartlett's three-sample methods, OLP and MA regression analysis.[11,12,21]

I start by examining the case for using techniques of regression that are non-parametric. These comprise the P-B and BLMS techniques (see Table S1). They are quite closely related and their authors claim that they are especially useful when there are outlying values.[25–27] However, they handle outlying values by more-or-less ignoring them in the process of constructing lines of best fit (see Figs 3,4).

These techniques are strongly advocated and used by clinical chemists (Tables 1 and S2), who seem to have a different attitude towards outliers than other biostatisticians, including me. Following the example of David Finney,[39] I have argued that the proper attitude is to discard outlying values only if they are due to a temporary malfunction of the measuring instrument, an aberration on the part of the measurer, a transcription error or impossible values.[29] Otherwise, they must surely be regarded as 'real' and caused by factors such as erratic behaviour of the measuring instrument, bias in the sampling process or even a non-linear rather than linear relationship between the measurements. For these reasons, I see no merit in the non-parametric approach.

This leaves for consideration by pharmacologists and physiologists only MA and OLP regressions. Under most circumstances, the two techniques give very similar regression lines (Figs 3,4). However, if the *x*,*y* measurements have been made on different scales or if the 'true' slope of the regression is far from $\beta = 1.0$ or $\beta = -1.0$, MA regression is unsuitable unless the data are standardized (SMA). But

Table 1 Thirteen Monte Carlo simulation studies of linear regression techniques for comparing methods of measurement, in order of publication

Discipline	Techniques studied	Monte Carlo features	Recommended techniques
Pharmacology[47]	OLS, MA, WMA, OLP, WLP +++	n = 6–48; β = 1.0; ρ = 0.88, 1.0	Undecided
Chemistry[42]	MA, Bartlett, Mandel	β = 0.90	MA, Mandel
Astronomy[48]	OLS, OLS bisector, MA, OLP	n = 20–500; β = 1.0, 2.0; ρ = 0.25–1.0	OLS bisector
Allometry[40]	MA, SMA (OLP)	n = 10, 20; ρ = 0.1–0.9	Log MA, OLP*
Chemistry[33]	OLS, MA, WMA	n = 50; β = 1.0	WMA
Chemistry[34]	OLS, WLS, MA, WMA, P-B	n = 50; β = 1.0	MA, WMA
Chemistry[22]	OLS, WMA, P-B	n = 40; ρ ≥ 0.96	P-B, WMA
Mathematics[49]	OLS, LAD, LAPD, TLS (MA), SHVD	n = 12–48; β = 1.0	SHVD, AHVD
Chemistry[27]	BLS, BLMS	n = 6, 90; β = 1.0	BLMS
Chemistry[50]	OLS, WLS, BLS	n = 5–100; β = 1.0	BLS
Chemistry[51]	EM, GLS, ODR, MM, CVR, OR (MA)	n = 51; β = 0.67	EM, MM
Biology[41]	MA, SMA (OLP)	n = 10–90; ρ = 0.5, 0.75	OLP
Geology[52]	OLS, WLS, MA, BLMS, EM, RR, RT ++	β = 1.0, 1.05	RR, RT, EM

*If $n < 20$, $r \geq 0.6$.

n, sample sizes; β, slope of Y on X for population in model $Y = \alpha + \beta X$; ρ, correlation coefficient for population; r, Pearson's correlation coefficient for sample; OLS, ordinary least squares regression; MA, major axis regression; WMA, weighted major axis regression; OLP, ordinary least products regression; WLP, weighted least products regression; Bartlett, Bartlett's 3 group method; Mandel, Mandel's method; SMA, standardized major axis regression; P-B, Passing–Bablok; LAD, least absolute deviations; LAPD, least absolute perpendicular deviations; TLS, total least squares; SHVD, squares of horizontal and vertical deviations; BLS, bivariate least squares; BLMS, bivariate least medians squared; WLS, weighted least squares; EM, expectation minimization; GLS, generalized least squares; ODR, orthogonal distance regression; MM, method of moments; CVR, constant variance ratio; OR, orthogonal regression; EM, expectation minimization; RR, Riu-Rius method; RT, Riply-Thompson method; WLP, weighted least products; +++, RT; ++, Ripley-Thompson; AHVD, absolute horizontal and vertical distances.

A description of the techniques is available in Table S1.

SMA regression is, in fact, identical to OLP regression. And if the scatter of y values increases as the x values increase (heteroscedasticity), a weighted version of MA (WMA) or OLP (WLP) should be used.

So, in summary, I recommend the following sequence when preparing to analyse a two-method comparison study by linear regression analysis:

1. Be clear why you plan to undertake linear regression analysis. Is it to calibrate one method against the other? Is it to detect fixed and/or proportional bias?
2. Proceed to construct a scattergram. Do the x,y values fall approximately on a straight line? Would they do so more clearly if you transformed the data (e.g. log transformation; see Fig. 4)? Are there outlying values? If so, are there sound reasons for deleting them (see Ludbrook[29])? Do you have evidence that there is a linear association between x and y values? Test this by calculating Pearson's product–moment correlation coefficient; if the value of r is small and if, for instance, it is not attended by $P \leq 0.05$, it is not worthwhile proceeding further.
3. Decide whether Model I or Model II regression analysis is appropriate. If the x values are fixed by the design of the study and can be regarded as error free, use Model I regression analysis (OLS); or, if there is heteroscedasticity, weighted least squares (WLS). If it would be absurd to predict x from y, rather than y from x, some would advise that Model I regression analysis can be used.
4. Otherwise, proceed to undertake OLP regression analysis or, if there is heteroscedasticity, WLP regression analysis. The regression coefficients can be used for calibrating one method against the other. If measurements were made on the same scale, the 95% CIs for the coefficients can be used to detect bias. If the 95% CI for a does not include 0, there is fixed bias. If the 95% CI for b does not include 1.0, there is proportional bias.
5. Even if there is no bias, you may still want to conclude that one method could, or could not, be substituted for the other. It may help you make this judgement if you construct the hyperbolic 95% prediction limits for the population represented by the OLP regression. This can be done by adapting the method for OLS regression.[12,43] If there is heteroscedasticity, V-shaped 95% confidence limits for OLP regression can be constructed by adapting the method of Bland.[15,44]
6. If you worry that the 95% prediction limits are widely spaced, perhaps you should have made replicate measurements by each method. This would allow you to compare the repeatability (measurement error) of each of the two methods.[6] It may even turn out that the standard method is attended by greater measurement error than the new method.

This article has been about comparing two sets of measurements. Several colleagues have asked, 'What if I want to intercompare more than two measurers or methods?' Carstensen has recently proposed a complex form of ANOVA for dealing with this situation,[45] but his proposed techniques of analysis are concerned with the Altman–Bland method of differences and he uses the same form of logic as Model I, least squares, regression analysis. It is beyond my capability to translate this approach into a Model II format. In such cases, it has been my practice to make multiple pairwise comparisons between the measurers or methods of measurement and, of course, to make an adjustment for testing multiple hypotheses.[46] There is another matter that I have not touched on. What if one wants to compare the slopes of two straight lines? The conventional approach is a complex ANOVA in which covariant analysis for slope is performed. But this assumes a Model I, least squares, approach to estimating the slopes, whereas, as I have argued, Model II (OLP) regression should be used. I cannot devise a way of transforming covariant analysis for slope to cater for OLP regression. My only solution is to declare two OLP slopes different if the 95% CIs for slope do not overlap.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Table S1 Some of the linear regression techniques used in method comparison studies
Table S2 Artificial data sets from Ludbrook (2008)
Table S3 Real-life data set from Rousseeuw and Leroy (2003), analyzed by Feldmann *et al.* (1998)
Appendix S1 Software for executing Model II regression analysis.
Appendix S2 Loss functions for Model II linear regressions.

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